5.3 - Eulers Theorems and Fleurys Algorithm

Problem Solving, Math 1000

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Question

How can we determine if a graph has an Euler path or an Euler circuit? How can we find such a circuit or a path if one exists?

Theorem (Euler's Circuit Theorem)

If a graph is connected and every vertex is even, then the graph has an Euler circuit. If a graph has any odd vertices, it does not have an Euler circuit.

Theorem (Euler's Path Theorem)

If a graph is connected and has exactly two odd vertices, then it has an Euler path. Any such path must start at one of the odd vertices and end at the other. If a graph has more than two odd vertices, it cannot have an Euler path.

Theorem (Euler's Sum of Degrees Theorem)

The sum of the degrees of all the vertices in a graph equals twice the number of edges. Therefore, the sum of the degrees of all the vertices in a graph is always even.

Algorithm (Fleury's Algorithm)

1. First, make sure your graph is connected and either has (1) no odd vertices (for a circuit), or (2) only two odd vertices (for a path).

2. Choose a starting vertex. In (1), it may be any vertex. In (2), it must be one of the two odd vertices.

3. Move down any non-bridge edge connected to your starting vertex if one is available. Otherwise, if no such edge is available, move across a bridge. Repeat this step, moving down only edges you have not previously traversed, until there are no remaining edges.